Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester IV)

AI(Artificial Intelligence)

Practical VIII

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Date of Assignment:22-08-2026	Date/Time of Submission:22-08-2026

Aim: Ethics and Bias Analysis

Select any publicly available dataset (for example, Iris, Adult Income, or a classification dataset of your choice) and analyze it for possible bias and class imbalance.

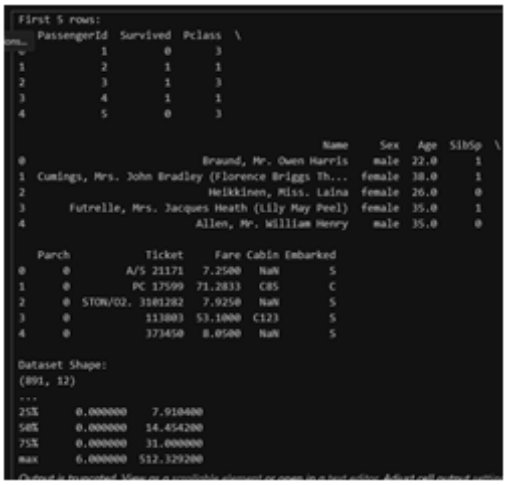
Dataset: Titanic train.csv from Kaggle

1. Load and Explore the Dataset

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv("titanic_train.csv")
print("First 5 rows:")
print(df.head())
print("\nDataset Shape:")
print(df.shape)
print("\nColumn Names:")
print(df.columns)
print("\nDataset Information:")
print(df.info())
print("\nStatistical Summary:")
print(df.describe())
```

Output:



2. Check Data Quality

Code:

```
print("Missing Values:")
print(df.isnull().sum())
print("\nDuplicate Rows:")
print(df.duplicated().sum())
print("\nData Types:")
print(df.dtypes)
```

Output:

```
Missing Values:
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

```
Duplicate Rows:
0

Data Types:
PassengerId      int64
Survived         int64
Pclass           int64
Name               str
Sex                str
Age            float64
...
Fare           float64
Cabin              str
Embarked           str
dtype: object
```

3. Identify the Target Variable

Code:

```
target = "Survived"
print("Target Variable:", target)
print("\nTarget Values:")
print(df[target].value_counts())
```

Output:

```
Target Variable: Survived

Target Values:
Survived
0    549
1    342
Name: count, dtype: int64
```

4. Analyze Class Distribution

Code:

```
print(df["Survived"].value_counts())
print("\nClass Distribution (%):")
print(df["Survived"].value_counts(normalize=True) * 100)
```

Output:

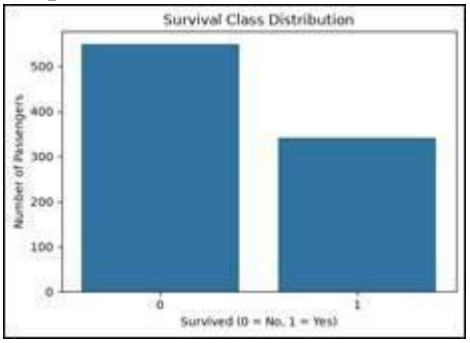
```
Survived
0    549
1    342
Name: count, dtype: int64

Class Distribution (%):
Survived
0    61.616162
1    38.383838
Name: proportion, dtype: float64
```

5. Visualize Class Imbalance

Code:

```
plt.figure(figsize=(6,4))
sns.countplot(x="Survived", data=df)
plt.title("Survival Class Distribution")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Number of Passengers")
plt.show()
```

Output:



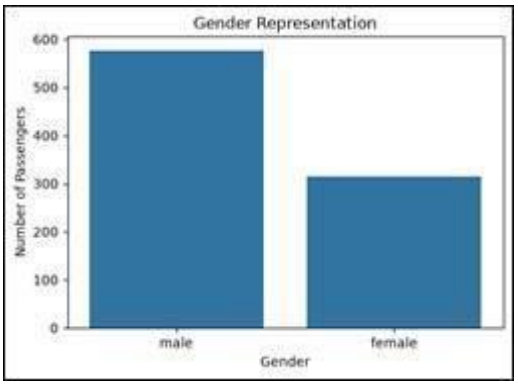
6. Analyze Gender Representation and display graph

Code:

```
print("Gender Representation:")
print(df["Sex"].value_counts())
plt.figure(figsize=(6,4))
sns.countplot(x="Sex", data=df)
plt.title("Gender Representation")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")
plt.show()
```

Output:

```
Gender Representation:
Sex
male      577
female    314
Name: count, dtype: int64
```



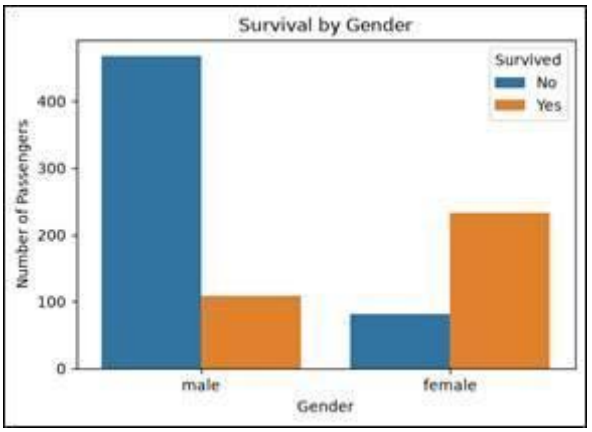
7. Analyze Survival by Gender and display graph

Code:

```
print("Survival by Gender:")
print(pd.crosstab(df["Sex"], df["Survived"]))
plt.figure(figsize=(6,4))
sns.countplot(x="Sex", hue="Survived", data=df)
plt.title("Survival by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")
plt.legend(title="Survived", labels=["No", "Yes"])
plt.show()
```

Output:

```
Survival by Gender:
Survived    0    1
Sex
female     81  233
male      468  109
```



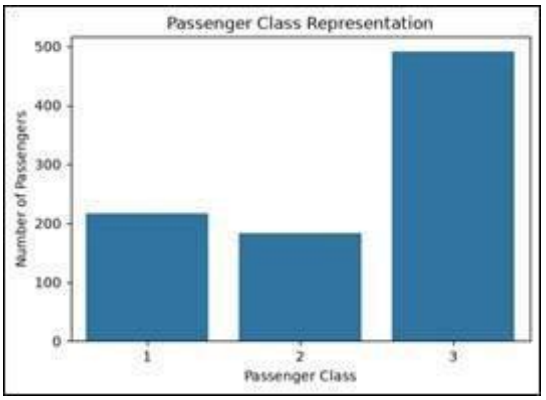
8. Analyze Passenger-Class Representation and display graph

Code:

```
print("Passenger Class Representation:")
print(df["Pclass"].value_counts().sort_index())
plt.figure(figsize=(6,4))
sns.countplot(x="Pclass", data=df)
plt.title("Passenger Class Representation")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")
plt.show()
```

Output:

```
Passenger Class Representation:
Pclass
1    216
2    184
3    491
Name: count, dtype: int64
```



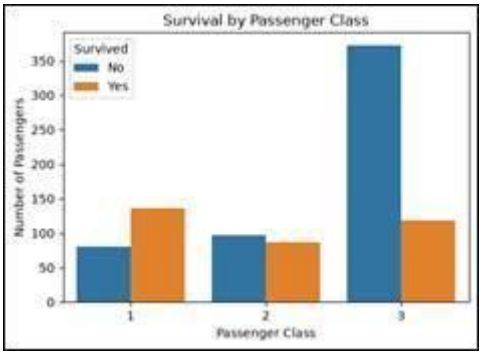
9. Analyze Survival by Passenger Class and display graph

Code:

```
print("\nCredit Risk by Age Group:")
print(pd.crosstab(df["AgeGroup"], df["kredit"]))
plt.figure(figsize=(8,4))
sns.countplot(data=df, x="AgeGroup", hue="kredit")
plt.title("Credit Risk by Age Group")
plt.xlabel("Age Group")
plt.ylabel("Number of Customers")
plt.xticks(rotation=20)
plt.show()
```

Output:

```
Survival by Passenger Class:
Survived    0    1
Pclass
1          80  136
2          97   87
3         372  119
```



10. Analyze Age Groups

Code:

```
df["AgeGroup"] = pd.cut(
    df["Age"],
          bins=[0, 12, 19, 59, 100],
    labels=["Child", "Teenager", "Adult", "Senior"]
)
print("Age Group Distribution:")
print(df["AgeGroup"].value_counts().sort_index())
```

Output:

```
Age Group Distribution:
AgeGroup
Child        69
Teenager     95
Adult       524
Senior       26
Name: count, dtype: int64
```

11. Analyze Survival by Age Group and display graph

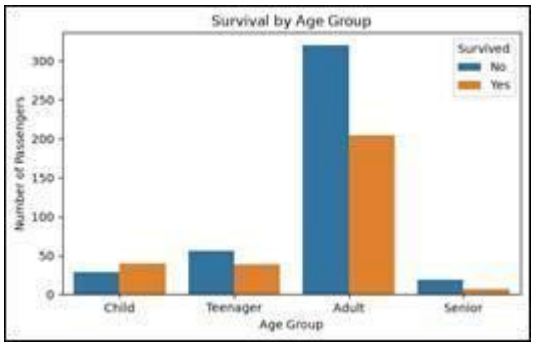
Code:

```
print("Survival by Age Group:")
print(pd.crosstab(df["AgeGroup"], df["Survived"]))
```

```
plt.figure(figsize=(7,4))
sns.countplot(
   x="AgeGroup",
   hue="Survived",
         data=df,
   order=["Child", "Teenager", "Adult", "Senior"]
)
plt.title("Survival by Age Group")
plt.xlabel("Age Group")
plt.ylabel("Number of Passengers")
plt.legend(title="Survived", labels=["No", "Yes"])
plt.show()
```

Output:

```
Survival by Age Group:
Survived    0    1
AgeGroup
Child      29   40
Teenager   56   39
Adult     320  204
Senior     19    7
```



12. Identify Possible Bias

Ans:

1. Gender bias may be present.
2. Passenger class may affect survival.
3. Age groups have different survival rates.
4. Missing data can also introduce bias.

13. Impact on ML

Ans:

1. Bias in data can make ML predictions less accurate or unfair.
2. The model may learn unwanted patterns from gender, age, or class.

14. Suggested solutions

Ans:

1. Handle missing values properly.
2. Balance the classes if required.
3. Remove duplicate or incorrect data.
4. Check model performance for different groups.

15. Final conclusion

Ans:

The Titanic dataset shows that gender, age, and passenger class affected survival. Proper data cleaning and bias handling are important for building a fair and reliable ML model.

Dataset: German Credit Risk Dataset from Kaggle

1. Load and Explore the Dataset:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
# Load Dataset
df = pd.read_csv("german_credit_data.csv")
# Remove spaces from column names
df.columns = df.columns.str.strip()
# 1. Load and Explore Dataset
print("First 5 Rows:")
print(df.head())
print("\nDataset Shape:")
print(df.shape)
print("\nColumn Names:")
```

```
print(df.columns.tolist())
print("\nDataset Information:")
print(df.info())
print("\nStatistical Summary:")
print(df.describe())
```

```
First 5 Rows:
   laufkont  laufzeit  moral  verw  hoehe  sparkont  beszeit  rate  famges  \
0         1        18      4     2   1049         1        2     4       2
1         1         9      4     0   2799         1        3     2       3
2         2        12      2     9    841         2        4     2       2
3         1        12      4     0   2122         1        3     3       3
4         1        12      4     0   2171         1        3     4       3

   buerge  ...  verm  alter  weitkred  wohn  bishkred  beruf  pers  telef  \
0       1  ...     2     21         3     1         1      3     2      1
1       1  ...     1     36         3     1         2      3     1      1
2       1  ...     1     23         3     1         1      2     2      1
3       1  ...     1     39         3     1         2      2     1      1
4       1  ...     2     38         1     2         2      2     2      1

   gastarb  kredit
0        2       1
1        2       1
2        2       1
3        1       1
4        1       1

[5 rows x 21 columns]
```

```
Dataset Shape:
(1000, 21)
```

```
Column Names:
['laufkont', 'laufzeit', 'moral', 'verw', 'hoehe', 'sparkont',
```

```
Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 21 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   laufkont  1000 non-null   int64
 1   laufzeit  1000 non-null   int64
 2   moral     1000 non-null   int64
 3   verw      1000 non-null   int64
 4   hoehe     1000 non-null   int64
 5   sparkont  1000 non-null   int64
 6   beszeit   1000 non-null   int64
 7   rate      1000 non-null   int64
 8   famges    1000 non-null   int64
...
```

2. Check Data Quality

```
# 2. Check Data Quality
print("\nMissing Values:")
print(df.isnull().sum())
print("\nDuplicate Rows:")
print(df.duplicated().sum())
```

3. Identify the Target Variable

```
# 3. Identify Target Variable
target = "kredit"
print("\nTarget Variable:", target)
```

```
kredit
1    70.0
0    30.0
```

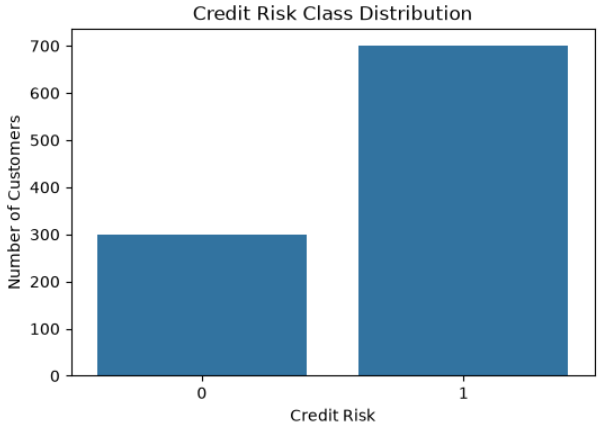
4. Analyze Class Distribution

```
# 4. Analyze Class Distribution
print("\nCredit Risk Distribution:")
print(df["kredit"].value_counts())
print("\nCredit Risk Percentage:")
print(df["kredit"].value_counts(normalize=True) * 100)
```

```
Name: proportion, dtype: float64
```

5. Visualize Class Imbalance

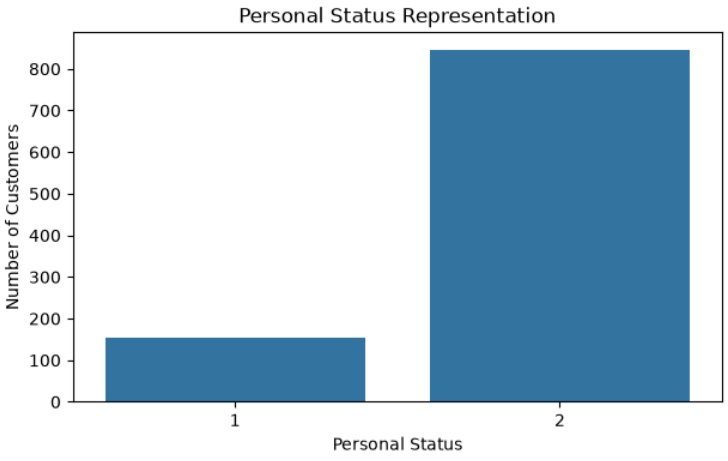
```
# 5. Visualize Class Imbalance
plt.figure(figsize=(6,4))
sns.countplot(data=df, x="kredit")
plt.title("Credit Risk Class Distribution")
plt.xlabel("Credit Risk")
plt.ylabel("Number of Customers")
plt.show()
```



6. Analyze Gender Representation and display graph

```
# 6. Analyze Gender / Personal Status Representation
print("\nPersonal Status Representation:")
print(df["pers"].value_counts())
plt.figure(figsize=(7,4))
sns.countplot(data=df, x="pers")
plt.title("Personal Status Representation")
plt.xlabel("Personal Status")
plt.ylabel("Number of Customers")
plt.show()
```

```
Personal Status Representation:
pers
2    845
1    155
Name: count, dtype: int64
```

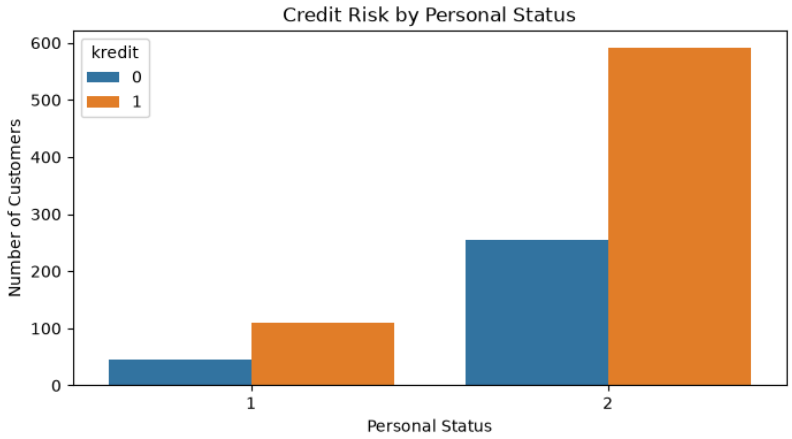


7. Analyze Credit Risk by Gender and display graph

```
# 7. Analyze Credit Risk by Personal Status
print("\nCredit Risk by Personal Status:")
print(pd.crosstab(df["pers"], df["kredit"]))
plt.figure(figsize=(8,4))
sns.countplot(data=df, x="pers", hue="kredit")
```

```
plt.title("Credit Risk by Personal Status")
plt.xlabel("Personal Status")
plt.ylabel("Number of Customers")
plt.show()
```

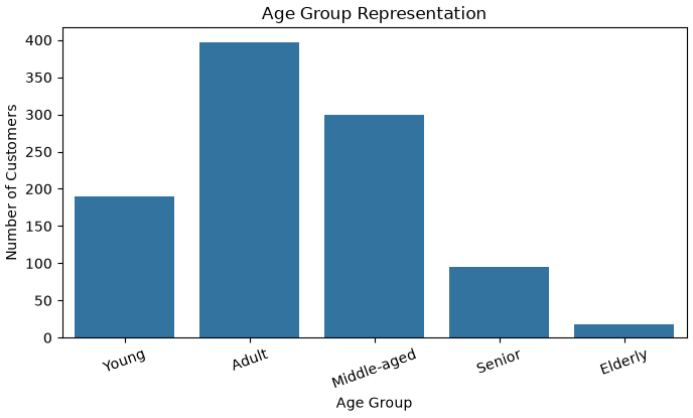
```
Credit Risk by Personal Status:
kredit    0    1
pers
1        46  109
2       254  591
```



8. Analyze Age Groups and display graph

```
# 8. Analyze Age Groups
df["AgeGroup"] = pd.cut(
    df["alter"],
    bins=[0, 25, 35, 50, 65, 100],
    labels=["Young", "Adult", "Middle-aged", "Senior", "Elderly"]
)
print("\nAge Group Representation:")
print(df["AgeGroup"].value_counts().sort_index())
plt.figure(figsize=(8,4))
sns.countplot(data=df, x="AgeGroup")
plt.title("Age Group Representation")
plt.xlabel("Age Group")
plt.ylabel("Number of Customers")
plt.xticks(rotation=20)
plt.show()
```

```
Age Group Representation:
AgeGroup
Young          190
Adult          397
Middle-aged    300
Senior          95
Elderly         18
Name: count, dtype: int64
```

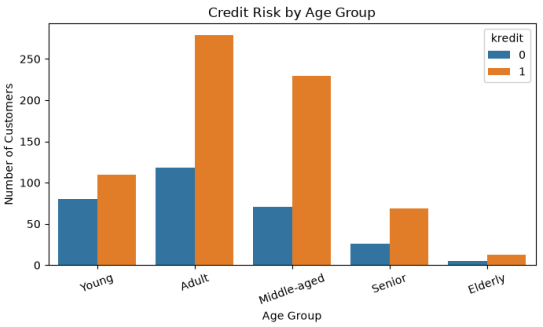


9. Analyze Credit Risk by Age Group and display graph

```
# 9. Analyze Credit Risk by Age Group
print("\nCredit Risk by Age Group:")
print(pd.crosstab(df["AgeGroup"], df["kredit"]))
plt.figure(figsize=(8,4))
```

```
sns.countplot(data=df, x="AgeGroup", hue="kredit")
plt.title("Credit Risk by Age Group")
plt.xlabel("Age Group")
plt.ylabel("Number of Customers")
plt.xticks(rotation=20)
plt.show()
```

```
Credit Risk by Age Group:
kredit          0    1
AgeGroup
Young          80  110
Adult         118  279
Middle-aged    71  229
Senior         26   69
Elderly         5   13
```



10. Analyze Housing Category Representation and display graph

```
# 10. Analyze Housing Category Representation
print("\nHousing Category Representation:")
print(df["wohn"].value_counts())
plt.figure(figsize=(7,4))
sns.countplot(data=df, x="wohn")
plt.title("Housing Category Representation")
plt.xlabel("Housing Category")
plt.ylabel("Number of Customers")
plt.show()
```

```
Housing Category Representation:
wohn
2    714
1    179
3    107
Name: count, dtype: int64
```

11. Analyze Credit Risk by Housing Category and display graph

```
# 11. Analyze Credit Risk by Housing Category
print("\nCredit Risk by Housing Category:")
print(pd.crosstab(df["wohn"], df["kredit"]))
plt.figure(figsize=(8,4))
sns.countplot(data=df, x="wohn", hue="kredit")
plt.title("Credit Risk by Housing Category")
plt.xlabel("Housing Category")
plt.ylabel("Number of Customers")
plt.show()
```

```
Credit Risk by Housing Category:
kredit    0    1
wohn
1        70  109
2       186  528
3        44   63
```

12. Identify Possible Bias

- Credit risk classes may be imbalanced.
- Some age, personal-status, or housing groups may be underrepresented.
- This can create unfair patterns in the data.

13. Impact on ML

- The model may favor the majority class.
- Predictions may be less accurate for underrepresented groups.
- Sensitive features may lead to biased decisions.

14. Suggested Solutions

- Balance the classes.
- Handle missing data properly.
- Use stratified sampling.
- Check model performance across different groups.

15. Final Conclusion

The dataset may contain imbalance and group-related bias. Proper preprocessing, balancing, and fairness checks can help build a more reliable ML model.